

ROYALE CONCORDE INTERNATIONAL SCHOOL

IP INVESTIGATORY PROJECT

Stock Market Portfolio Builder (SMPB)



BONAFIDE CERTIFICATE

This is to certify that _____ has satisfactorily completed the project in Chemistry prescribed by the CBSE AISSCE course in the laboratory of Royale Concorde International School in the year 2022-2023.

Signature of
External Examiner

Signature of
Internal Examiner

Date:

Name of the Candidate:

Roll No:

School Seal:

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to the people who have helped and supported me throughout the project

I express my sincere gratitude to the principal Mrs.Pinkey Singh for providing us with an environment to complete our project successfully

I extend my deepest thanks to my Informatics Practices Teacher Mrs.Saswati Sarangi for their continuous and unwavering support during the entire course of this project and make necessary correction whenever required

I would like to thank my project partner, family and friends for their constant support through the project .

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PROJECT DESCRIPTION

Since many people globally, especially the people in their early 20's, lack basic financial knowledge, it becomes a major barrier for people who want to invest in the stock market. This programme will suggest stocks to people according to the amount specified by them according to past market data. This initial step helps them largely by showing them a main starting point upon which specific research into the suggested stocks can be made by the user and then money can accordingly be invested.

Hence, we have designed a program to analyze the data of the past 5 years' stock market and suggest stocks according to the investment to be made.

HARDWARE REQUIREMENTS

A Computer/Laptop with
Operating System-Windows 7 or MacOS
x86 64-bit CPU (Intel / AMD architecture)
4 GB RAM.
2 GB free disk space.

SOFTWARE REQUIREMENTS

Python 3.6.x or higher version
Pandas Library preinspandas_datareader
Yahoo_fin
datetime
timetalled
Matplotlib Library preinstalled

CODE

Import statements:

```
import datetime as dt
import os
from tkinter import Tk
from tkinter.filedialog import askopenfilename
import matplotlib.pyplot as plt
import pandas as pd
import yfinance as yf
from pandas import ExcelWriter
from pandas_datareader import data as pdr
```

Initialization

```
yf.pdr_override()
start=dt.datetime(2010,12,1)
now = dt.datetime.now()

root = Tk()
ftypes = [(".xlsm","*.xlsx",".xls")]
ttl = "Title"
dir1 = 'C:\\'
filePath = askopenfilename(filetypes = ftypes, initialdir = dir1,
title = ttl)
#filePath=r"C:\Users\MANAV\Desktop\stock
screener\National_Stock_Exchange_of_India_Ltd.xlsx"

stocklist = pd.read_excel(filePath)
#stocklist=stocklist.head()
```

```
exportList= pd.DataFrame(columns=['Stock', "50 Day MA", "150 Day Ma",  
"200 Day MA", "52 Week Low", "52 week High"])
```

```
for i in stocklist.index:
```

```
    stock=str(stocklist["Symbol"][i])
```

```
    try:
```

```
        df = pdr.get_data_yahoo(f'{stock}.NS', start, now)
```

```
        smaUsed=[50,150,200]
```

```
        for x in smaUsed:
```

```
            sma=x
```

```
            df['SMA_' + str(sma)] =
```

```
round(df.Close.rolling(window = sma).mean())
```

```
currentClose=df["Adj Close"][-1]
```

```
    moving_average_50=df["SMA_50"][-1]
```

```
    moving_average_150=df["SMA_150"][-1]
```

```
    moving_average_200=df["SMA_200"][-1]
```

```
    low_of_52week=min(df["Adj Close"][-260:])
```

```
    high_of_52week=max(df["Adj Close"][-260:])
```

```
    try:
```

```
        moving_average_200_20 = df["SMA_200"][-20]
```

```
    except Exception:
```

```
        moving_average_200_20=0
```


Condition 1: Current Price > 150 SMA and > 200 SMA

```
if(currentClose>moving_average_150>moving_average_200):  
    cond_1=True  
else:  
    cond_1=False
```

Condition 2: 150 SMA and > 200 SMA

```
if(moving_average_150>moving_average_200):  
    cond_2=True  
else:  
    cond_2=False
```

Condition 3: 200 SMA trending up for at least 1 month (ideally 4-5 months)

```
if(moving_average_200>moving_average_200_20):  
    cond_3=True  
else:  
    cond_3=False
```

Condition 4: 50 SMA> 150 SMA and 50 SMA> 200 SMA

```
if(moving_average_50>moving_average_150>moving_average_200):  
    #print("Condition 4 met")  
    cond_4=True  
else:  
    #print("Condition 4 not met")  
    cond_4=False
```

Condition 5: Current Price > 50 SMA

```
if(currentClose>moving_average_50):
    cond_5=True
else:
    cond_5=False
```

Condition 6: Current Price is at least 30% above 52 week low (Many of the best are up 100-300% before coming out of consolidation)

```
if(currentClose>=(1.3*low_of_52week)):
    cond_6=True
else:
    cond_6=False
```

Condition 7: Current Price is within 25% of 52 week high

```
if(currentClose>=(.75*high_of_52week)):
    cond_7=True
else:
    cond_7=False
```

```
if(cond_1 and cond_2 and cond_3 and cond_4 and cond_5 and cond_6 and
cond_7):
```

```
    exportList = exportList.append({'Stock': stock, "50
Day MA": moving_average_50, "150 Day Ma": moving_average_150, "200
Day MA": moving_average_200, "52 Week Low": low_of_52week, "52 week
High": high_of_52week}, ignore_index=True)
except Exception:
    print("No data on "+stock)
```

```

print(exportList)
currenttime = str(dt.datetime.now().date()) + '_' +
str(dt.datetime.now().time()).replace('.', '-').replace(":", "-")
os.makedirs(os.path.dirname(filePath)+f"/{currenttime}")

for i in exportList.index:
    stock=str(exportList["Stock"][i])
    # print(stock)
    # print(i)

    try:
        df = pdr.get_data_yahoo(f'{stock}.NS', start, now)

```

Algorithm to calculate the moving averages:

```

smaUsed=[50,150,200]
    for x in smaUsed:
        sma=x
        df['SMA_' + str(sma)] =
round(df.Close.rolling(window = sma).mean())

currentClose=df["Adj Close"][-1]
    moving_average_50=df["SMA_50"][-1]
    moving_average_150=df["SMA_150"][-1]
    moving_average_200=df["SMA_200"][-1]
    low_of_52week=min(df["Adj Close"][-260:])
    high_of_52week=max(df["Adj Close"][-260:])

    plt.figure(figsize=(12,6))
    plt.plot(df.Close, label="Closing Price", color="green")
    plt.plot(df.SMA_150, label="SMA_150", color="b")
    plt.plot(df.SMA_200, label="SMA_200", color="r")
    plt.legend()

```

```
plt.xlabel("YEARS")
plt.ylabel("PRICE")
plt.title(stock)

plt.savefig(f'{os.path.dirname(filePath)}/{currenttime}/{stock}_{currenttime}')

except Exception as e:
    print(e)
```

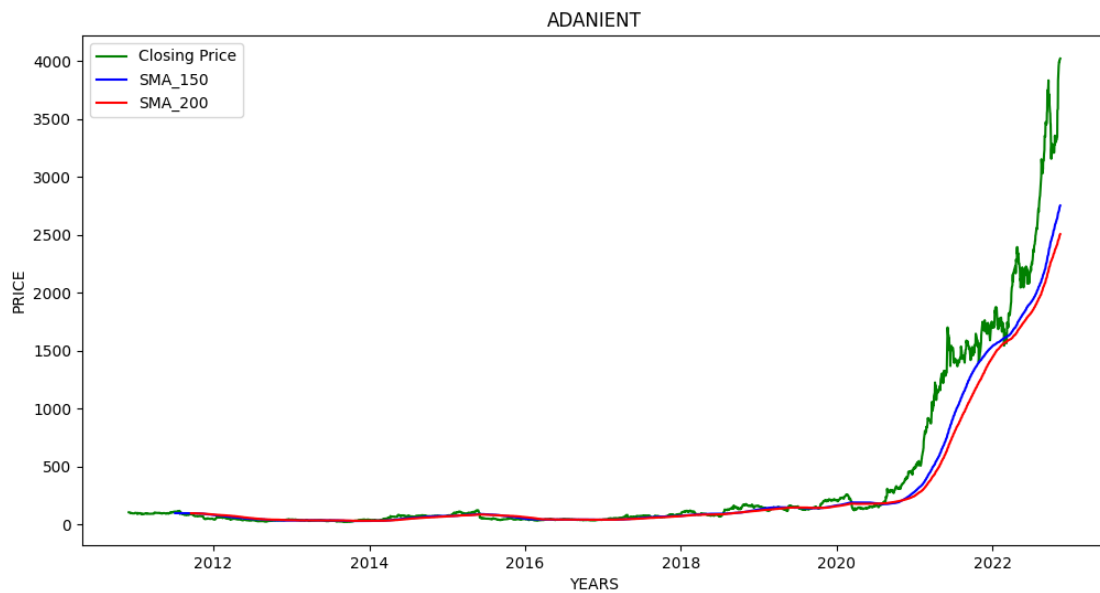
Saves output file in desktop:

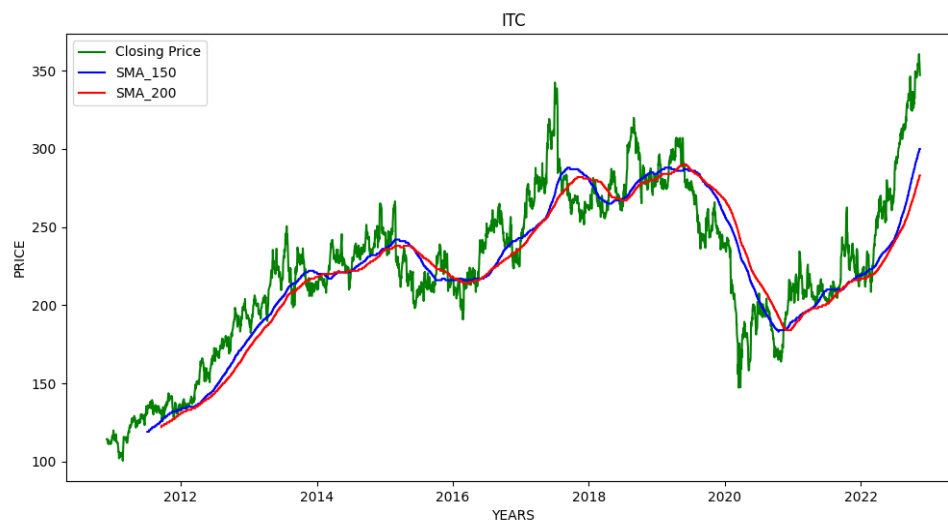
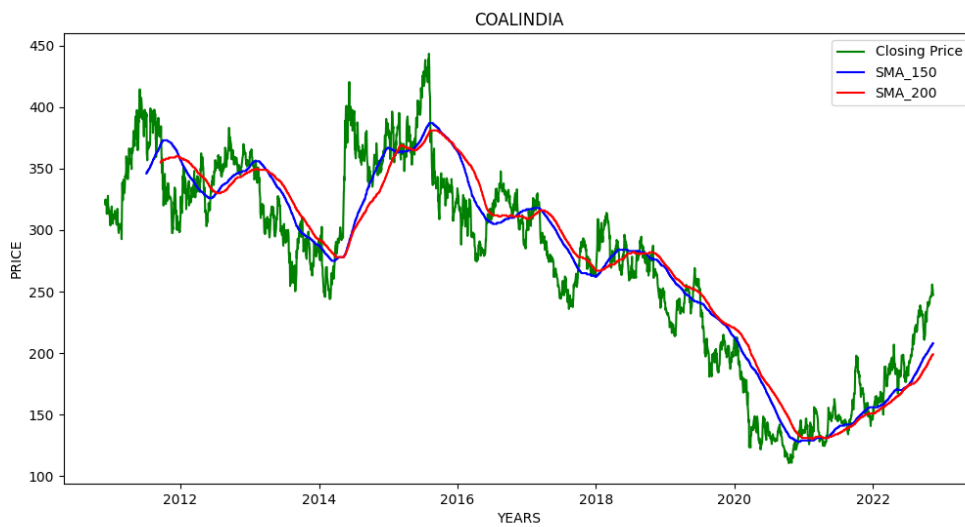
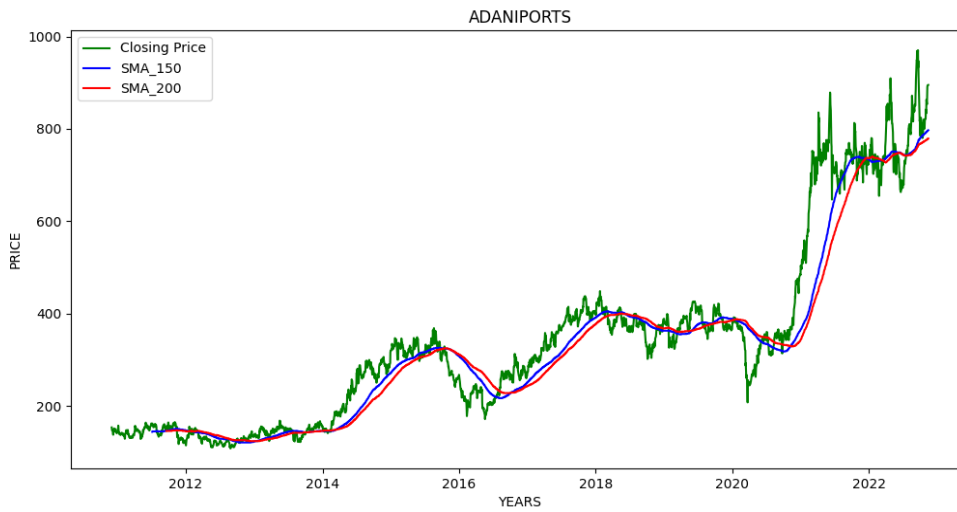
```
newFile =
f'{os.path.dirname(filePath)}/{currenttime}/ScreenOutput.xlsx'

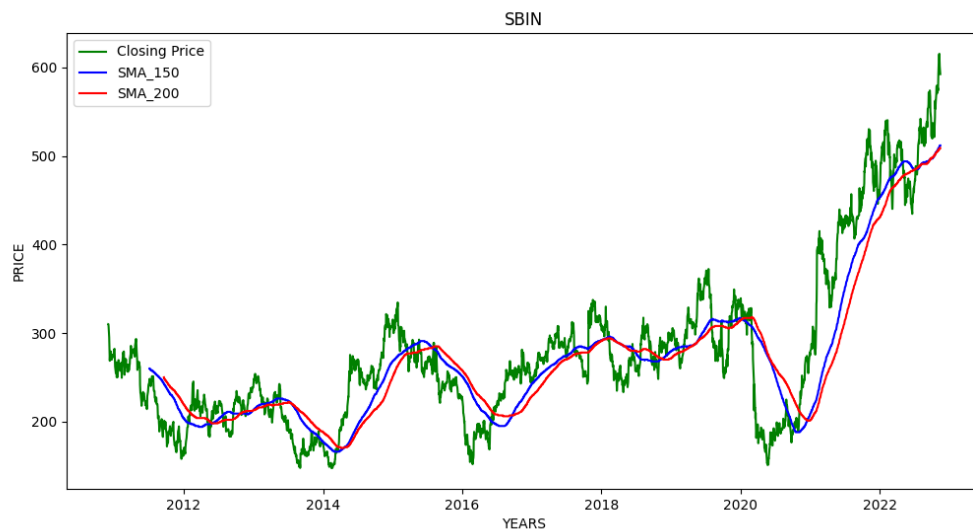
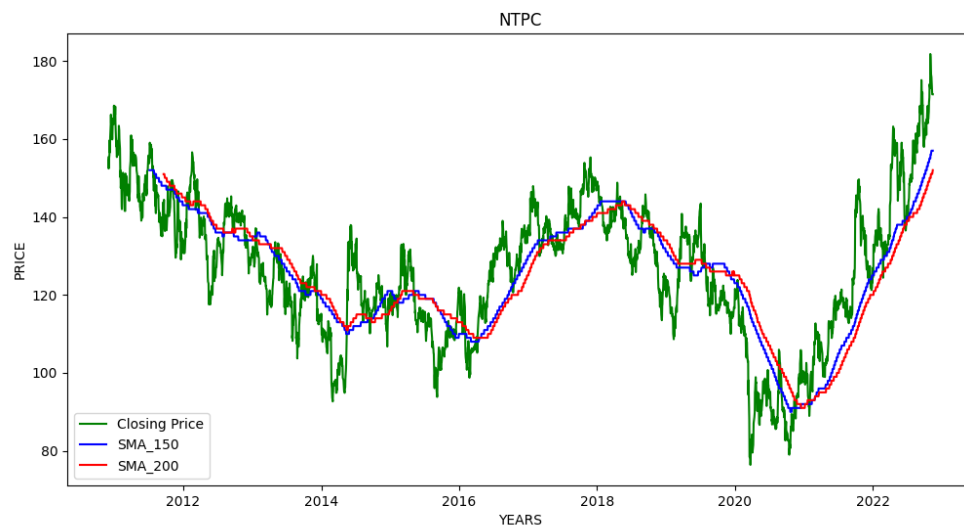
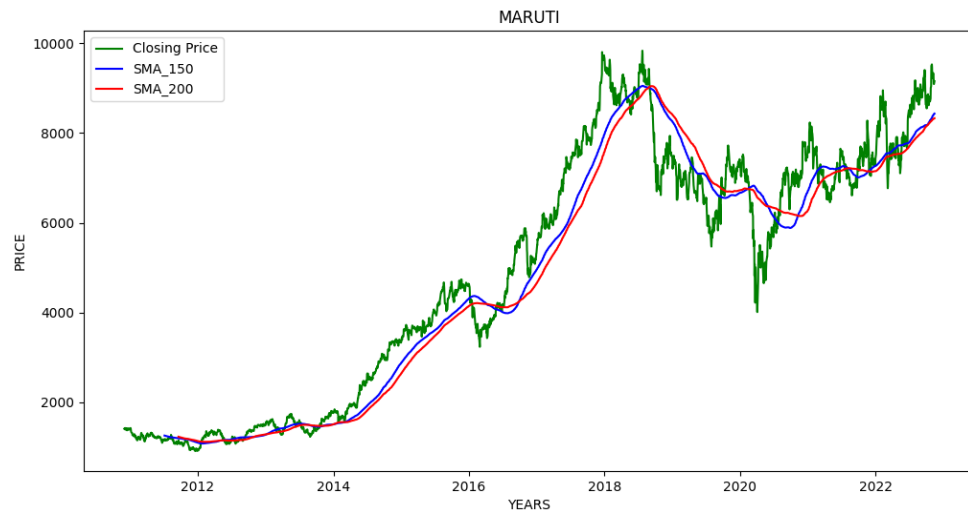
writer= ExcelWriter(newFile)
exportList.to_excel(writer, "Sheet1")
writer.save()
```

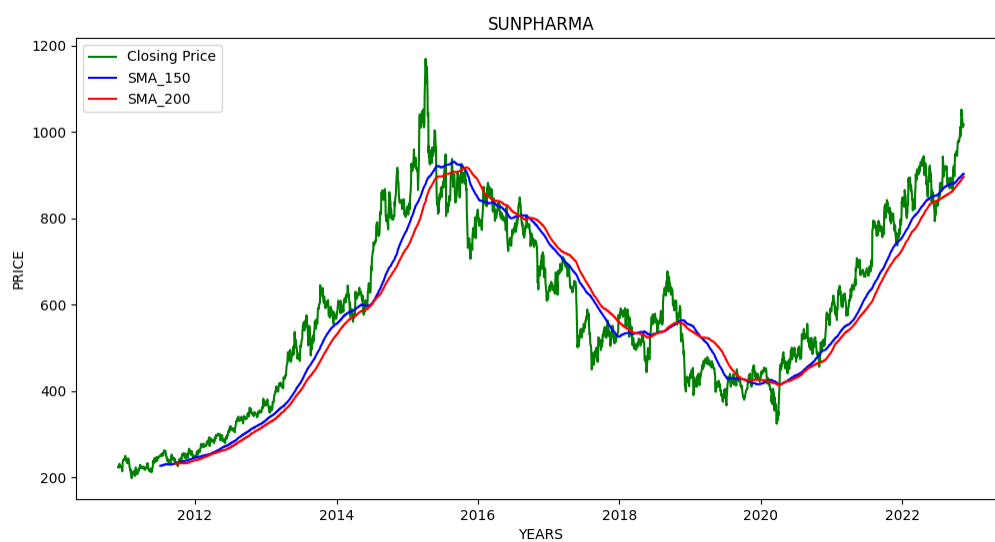
OUTPUT

	A	B	C	D	E	F	G	H
1		Stock	50 Day MA	150 Day Ma	200 Day MA	52 Week Low	52 week High	
2	0	ADANIENT	3496	2753	2506	1397.955811	4022.550049	
3	1	ADANIPORTS	856	797	779	650.3446655	970.25	
4	2	BRITANNIA	3789	3619	3555	2995.101318	4175.350098	
5	3	CIPLA	1103	1018	1006	856.0412598	1173.400024	
6	4	COALINDIA	234	208	199	134.6324158	255.8000031	
7	5	ICICIBANK	892	804	789	649.8502808	926.2999878	
8	6	ITC	338	300	283	202.2717896	360.7000122	
9	7	MARUTI	8974	8431	8328	6724.773926	9527.599609	
10	8	NTPC	168	157	152	115.596611	181.8000031	
11	9	SBIN	557	512	509	433.5342102	615.5499878	
12	10	SUNPHARMA	949	903	896	728.9854126	1051.949951	
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Bibliography

- > Source of csv File - Historical stock data for all current S&P 500 companies.
- > Online Python documentation - for python command syntax .
- > Online Javascript documentation -for javascript command syntax.
- > Text book - Class XI and XII - Informatics Practices NCERT.
- > Google.com - For any online queries.
- > Online articles related to our topic.